

The activity of metals decreases from top to bottom. The metals/non-metals on the bottom (with high positive standard reduction potentials) have the tendency to accept electrons readily, act as suitable cathodes/oxidizing agents. These are called least active or noble metals.

For example, in a Daniel cell, Zn electrode ($E^\circ = -0.76\text{V}$) acts as anode and undergo oxidation, whereas the Cu electrode ($E^\circ = 0.34\text{V}$) acts as cathode and undergo reduction.

Table 1.1 Electrochemical Series

Element	Electrode reaction (reduction)	E° (in volts)
Li	$\text{Li}^+ + \text{e}^- \longrightarrow \text{Li}$	-3.05
K	$\text{K}^+ + \text{e}^- \longrightarrow \text{K}$	-2.93
Na	$\text{Na}^+ + \text{e}^- \longrightarrow \text{Na}$	-2.71
Mg	$\text{Mg}^{2+} + 2\text{e}^- \longrightarrow \text{Mg}$	-2.37
Al	$\text{Al}^{3+} + 3\text{e}^- \longrightarrow \text{Al}$	-1.66
Zn	$\text{Zn}^{2+} + 2\text{e}^- \longrightarrow \text{Zn}$	-0.76
Fe	$\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe}$	-0.44
Cd	$\text{Cd}^{2+} + 2\text{e}^- \longrightarrow \text{Cd}$	-0.40
Ni	$\text{Ni}^{2+} + 2\text{e}^- \longrightarrow \text{Ni}$	-0.25
Sn	$\text{Sn}^{2+} + 2\text{e}^- \longrightarrow \text{Sn}$	-0.14
Pb	$\text{Pb}^{2+} + 2\text{e}^- \longrightarrow \text{Pb}$	-0.13
H	$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$	0.00
Cu	$\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$	0.34
Fe	$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$	0.77
Ag	$\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$	0.80
Pt	$\text{Pt}^{2+} + 2\text{e}^- \longrightarrow \text{Pt}$	1.20
Cl	$\text{Cl}_2 + 2\text{e}^- \longrightarrow 2\text{Cl}^-$	1.36
Au	$\text{Au}^{3+} + 3\text{e}^- \longrightarrow \text{Au}$	1.50